

$X = \begin{cases} 1 & \text{event ON} \\ 0 & \text{event OFF} \end{cases} \quad p \rightarrow \text{indicator variable}$

$$E[X] = p = \Pr(X=1)$$

buy headphones

n headphones $1, 2, \dots, n \rightarrow n=9$
 $\uparrow \quad \uparrow$
 worst best
 $1 \ 2 \ 3 \ \dots \ 9$
 $3 \ 3 \ 9 \ \dots \ 8 \rightarrow \text{How good is} \rightarrow \text{Cost} = 1$
 $3 \ 9 \ 2 \ 5 \ 6 \ 1 \ 7 \ 8 \ 9 \rightarrow \text{Cost} = 7$

random choose $\Rightarrow n!$ $\rightarrow \left\{ \begin{array}{l} 3456 \dots \\ 1247 \dots \\ \dots \end{array} \right\} \frac{\{\text{Costs}\}}{n!} = E(\text{Cost})$

$$H = [2, 4, 3, 5, 1, 7, 8, 6, 9]$$

$$X_i = \begin{cases} 1 & \text{if } i \text{ is chosen} \\ 0 & \end{cases} \quad X_1 = X_2 = X_3 = X_4 = X_5 = X_6 = X_7 = X_8 = X_9 = 1$$

$$\text{Cost} = \sum_{i=1}^n X_i$$

$$E[\text{Cost}] = E\left[\sum_{i=1}^n X_i\right] \xrightarrow{\text{linearity of } E[\cdot]} \sum_{i=1}^n E[X_i] = \sum_{i=1}^n \Pr(X_i=1) = \sum_{i=1}^n \frac{1}{i} \sim \ln n$$

harmonic sum

$$\Pr(X_i=1) = \frac{(i-1)!}{i!} = \frac{1}{i}$$

$$H = [1, 2, 3, \dots, i, \dots, n]$$

i ON $\nleftrightarrow H[i] > H[j]$ for $1 \leq j < i$

Given array A of n distinct elements, permute them uniformly of random (shuffling)

$\text{random}(a, b) \rightarrow x \in [a, \dots, b]$ uniformity of random

for $i=1 \dots n-1$:

$S = \text{random}(i \dots n)$

Swap($A[i]$, $A[S]$)

Use indicator variables to show that

this for loop gives one of $n!$ with $\text{pr} = \frac{1}{n!}$

Random QuickSort

Choose pivot uniformly or random among the n elements of A

random algo $\rightarrow$ expected cost $\rightarrow$ Prob space are all random choices of the pivot position in A

deterministic algo $\rightarrow$ expected cost $\rightarrow$ Prob space are all possible A 's

A sorted $z_1 < z_2 < \dots < z_n$

$$X_{i,j} = \begin{cases} 1 & z_i \in z_j \text{ compared} \\ 0 & \text{else} \end{cases}$$

Facts:

1) $z_i \in z_j$ compared $\Rightarrow$ one of them is a pivot

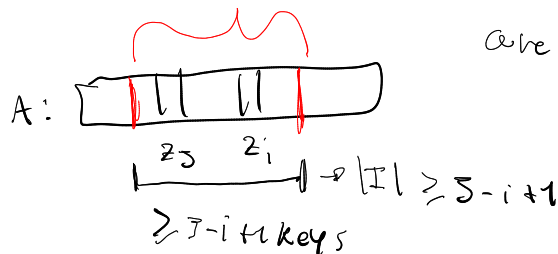
2) $z_i \in z_j$ compared at most once

$\sum_{1 \leq i < j \leq n} X_{i,j}$ = total number of comparison

made by Random QS

Recursive Call $\Rightarrow z_i, z_{i+1}, \dots, z_{j-1}, z_j$ are inside this call

3) $z_i \in z_j$ compared



expected cost

$$E \left[\sum_{i < j} X_{i,j} \right] = \sum_{i < j} E[X_{i,j}] = \sum_{i < j} \Pr(X_{i,j} = 1) = \sum_{i < j} \frac{2}{j-i+1} = \sum_{i=1}^n \sum_{j=i+1}^n \frac{2}{j-i+1} \approx O(n \log n)$$

Harmonic
 $O(\log n)$

$$\Pr(X_{i,j} = 1) = \Pr(z_i, z_j \in \text{same } I \text{ and } z_i \text{ or } z_j \text{ is a random pivot chosen in } I)$$

$$\leq \Pr(z_i \text{ or } z_j \text{ random pivot chosen in } I)$$

$$\leq \frac{2}{|I|} \leq \frac{2}{j-i+1}$$